

KARNATAKA RESIDENTIAL EDUCATIONAL INSTITUTIONS SOCIETY

ICT Class 8 - Unit Test 2

Unit Test 2 - Answer Key

Class: Class 8

Assessment Type: Unit Test 2

Total Marks: 10

Duration: 30 minutes

Topics Covered: Data Structures - Lists, Tuples, Dictionaries

ICT Teaching Resources Portal

Answer Key - Grading Guidelines

Note to Teacher: This answer key provides expected answers and marking guidelines. Accept equivalent answers where appropriate. Partial marks may be awarded for incomplete but correct responses.

Section I: Multiple Choice Questions (5 x 1 = 5 marks)

Q.No	Answer	Explanation
1	a	<code>pop()</code> removes and returns the last element (or specified index)
2	c	<code>TypeError</code> because tuples are immutable and cannot be modified
3	c	Dictionary keys must be immutable (strings, numbers, tuples)
4	c	<code>dict.keys()</code> returns a view object of all keys in the dictionary
5	a	<code>[x**2 for x in range(1,6)]</code> correctly creates list of squares

Section II: Short Answer Questions (5 x 1 = 5 marks)

1. Write the output and explain what happened.

- **Answer:** Output: [10, 40, 50]. The `del nums[1:3]` statement deleted elements at indices 1 and 2 (values 20 and 30) from the list.
- **Marking:** 1 mark for correct output, 1 mark for explanation.

2. Write a Python program to merge two dictionaries.

- **Answer:**

```
d1 = {"a": 1, "b": 2}
d2 = {"c": 3, "d": 4}

# Method 1: Using update()
merged = d1.copy()
merged.update(d2)

# Method 2: Using unpacking (Python 3.5+)
# merged = {**d1, **d2}

print(merged)
```

- **Marking:** 1 mark for correct dictionary creation, 1 mark for correct merge operation.

3. What will be the output and why?

- **Answer:** Output: 2 4 6. The loop iterates through each row in the nested list, and `row[1]` accesses the second element of each inner list.
- **Marking:** 1 mark for correct output, 1 mark for explanation.

4. Differentiate between `list.remove(x)` and `list.pop(index)`.

- **Answer:** `remove(x)` searches for the first occurrence of value `x` and removes it; returns nothing. `pop(index)` removes and returns the element at specified index; if no index given, removes last element.
- **Marking:** 1 mark for explaining `remove`, 1 mark for explaining `pop`.

5. Write a program to count character frequency.

- **Answer:**

```
def char_frequency(text):
    freq = {}
    for char in text:
        freq[char] = freq.get(char, 0) + 1
    # Sort by frequency
    sorted_freq = sorted(freq.items(), key=lambda x: x[1], reverse=True)
    return sorted_freq

text = "hello world"
for char, count in char_frequency(text):
    print(f"'{char}': {count}")
```

- **Marking:** 1 mark for dictionary creation, 1 mark for frequency counting and sorting.